BuildO

FULL FUNCTIONAL PRODUCT & DEVELOPER HANDOVER

Construction Ecosystem Platform • Ahenam Technologies

PEOPLE • WORK • PROJECTS • MATERIALS • RENTAL • TRANSPORT • MONEY

Purpose: This document is the functional blueprint for engineers. It describes how BuildO should behave, not just what modules exist.

Important: BuildO is not a labour-only app. Workforce is one module inside a project-centric construction operating network.

UI reference: Included mockups are visual direction; final UI should be refined by the engineering/design team without changing the functional flows.

1. Product Vision & Core Architecture

BuildO connects the construction ecosystem around a central Project. A project can consume people, work, materials, rental assets and transport, while every financial event is recorded in the Money ledger.

Core relationship: User/Company → Role → Project → Operational transaction → Money transaction → Audit trail.

The system must support marketplace activity and private company operations at the same time. A person may be a Labour, Contractor, Supplier, Rental Owner and/or Driver.

Core entity	Must link to
User	Roles, identity, permissions, notifications, financial account
Company	Users, projects, suppliers, inventory, roles, money
Project	Workers, jobs, materials, rentals, transport, payments, progress
Order	Buyer, supplier, products, project, transport, payment
Rental contract	Owner, renter, assets, project/site, dates, payment, return
Transport trip	Driver, vehicle, pickup, delivery, order, project, payment
Money transaction	Payer, recipient, project/order/rental/trip, method, status, audit

2. Account, Identity & Role System

Onboarding must ask what the user wants to do, then allow additional roles later. Do not create separate accounts for every business role.

Login → OTP/verified identity → role selection → role-specific onboarding → dashboard.

A user can switch roles from the profile/menu. Switching role changes dashboard, permissions and visible business context.

- Labour onboarding: personal details, phone, trade, skills, experience, availability, profile photo and optional documents.
- Company onboarding: company name, business type, registration/contact details, locations and staff/owner roles.
- Supplier onboarding: store details, delivery areas, categories and verification/KYC where applicable.
- Rental onboarding: owner/business details, asset categories, service area and verification.
- Transport onboarding: driver details, licence/verification, vehicle details, capacity and service area.
- Client/project owner onboarding: project/client profile and contact details.

Permission principle

Every API endpoint must check role + company/project relationship + action permission. Hiding a button in UI is not a security control.

3. Role-Based Dashboards

Dashboard is not shared across roles. Each role gets a purpose-built home screen with the most important actions and alerts for that role.

Role	Primary cards/actions
Labour	Today’s work, Check-in, jobs, requests, employer, earnings, attendance history, notifications
Contractor/Owner	Projects, workforce today, attendance, hire, material order, rental, transport, money, alerts
Supervisor	Assigned sites, today’s roster, attendance, tasks, progress, material requests, issues
Supplier	New orders, pending orders, catalogue, stock, quotations, deliveries, revenue
Rental Owner	Total assets, available, on-rental, reserved, maintenance, returns, locations, revenue
Driver/Transport	Available trips, accepted trips, pickup, delivery, navigation, status, earnings
Client	Projects, progress, orders, deliveries, payments, documents, approvals

4. Labour Functional Flow

Goal: help a worker find work, maintain a professional profile, attend assigned sites and receive money while keeping private employer information private.

Worker flow: Profile → Search Jobs/Requests → Apply/Accept → Company Approval → Assignment → GPS Attendance → Work History → Payment.

Joining/hiring

- Owner/Supervisor enters labour phone number → BuildO searches matching profile.
- Owner/Supervisor sends join request → labour receives notification.
- Labour accepts/rejects → company retains final approval.
- Labour may have multiple work requests but active employment/site assignments must respect company rules.

Attendance

- Assigned labour opens attendance → GPS/geofence check.
- If inside permitted site radius → Check-in enabled.
- Outside radius → check-in blocked with explanation, unless Supervisor/Owner marks attendance.
- Check-out records time; system calculates total working time.
- Overtime is calculated from configured shift/rules and can be edited only by authorized roles.
- Every edit records who changed it, old value, new value and timestamp.

Privacy

Labour must not see private employer-to-employer ratings/comments. Authorized owners can see private information subject to company permissions.

5. Work Hiring / Jobs

Jobs are a marketplace object that can be attached to a project or offered as standalone work.

Action	Function
Post job	Trade, quantity/headcount, location/site, start date, duration, wage/rate, requirements, description, photos
Discover	Filter by trade, distance, availability, date, temporary/permanent and rate
Apply	Labour sends application/request
Invite	Company invites selected labour
Accept/Reject	Labour responds
Approve	Company confirms hiring/assignment
Complete	Work record closes and becomes history + payment basis

6. Project Management

Project is the main operational container. Every project should have an overview and tabs/sections for People, Work, Materials, Rental, Transport, Money, Progress, Documents and Reports.

Project creation fields

- Project name, client, location/site, contract/project value, target margin, start date, planned end date.
- Stages with planned start/end and progress weight.
- Assigned owner/manager/supervisor.
- Budget/cost categories.
- Optional BOQ/material requirements and labour plan.

Progress engine

- Each stage has planned %, actual %, dates and status.
- Daily updates can include completed work, labour count, material usage, photos and remarks.
- System compares planned vs actual progress and planned vs actual cost.
- AI can alert on delays, abnormal costs or missing inputs.

7. Supplier & Material Marketplace

Supplier is a business role with a digital catalogue. Supplier controls what it can supply, pricing and stock.

Product creation

- Category, product name, brand, specification/size, unit, price, available stock, minimum order, delivery area, photos, description.
- Allow bulk products and variants (e.g., cement 50kg, steel 8mm, 10mm, etc.).
- Stock can be updated manually or automatically from orders.

Purchase flow

Step	System behaviour
1. Search	Buyer searches/filter products and suppliers.
2. Add to order	Select quantity, project/site and delivery preference.
3. Order	Supplier receives order; stock is reserved according to policy.
4. Accept	Supplier confirms price/availability and expected readiness.
5. Prepare	Supplier prepares goods.
6. Ready	Supplier taps Ready for Pickup; transport workflow is triggered.
7. Transport	Available eligible vehicles/drivers receive alert.
8. Delivery	Trip status updates; site receives delivery.
9. Receipt	Authorized recipient confirms received quantity/condition.
10. Close	Order, stock, project cost and Money ledger update.

8. Transport & Logistics

Transport is a network layer connecting suppliers/material orders to the destination site.

- Vehicle profile: type, registration, capacity, owner, driver, service area, availability.
- Supplier marks an order Ready for Pickup.
- BuildO finds eligible available vehicles based on location/capacity/service area.
- Driver receives request → views pickup, delivery, load, fee → Accept/Reject.
- Accept creates assignment and notifies supplier + buyer.
- Statuses: Assigned → Arrived Pickup → Loading → Picked Up → In Transit → Arrived Site → Delivered.
- GPS tracking may be enabled for active trips where permission/device support exists.
- Delivery confirmation updates order and Money.

9. Rental Inventory & Custody

Rental must maintain a real inventory ledger. It must answer how many assets the owner has, where they are and who has them.

Inventory state	Meaning
Available	Physically available for new rental
Reserved	Booked but not yet handed over
On Rental	Currently with customer/site
Maintenance/Damaged	Unavailable due to maintenance/damage
Lost/Issue	Exception requiring investigation

Bulk example

Owner has 1,000 jacks. 600 are rented: Available = 400; On Rental = 600. BuildO must show the 600 split by customer/site and expected return date.

Rental flow

- Create inventory → publish rental item/rate → customer requests dates/quantity → owner accepts → reserve quantity → handover → On Rental.
- Every handover records quantity, customer, project/site, date, expected return and condition.
- Return records quantity returned and condition. Available quantity increases automatically.
- Damaged quantity moves to Maintenance/Damaged and is excluded from available stock.
- Overdue returns create alerts.
- Individual high-value assets should have unique asset ID/QR code and full history.

10. Money Management & Payment Ledger

Money is a core module. Every role sees the financial information permitted for that role.

Overview cards

Card	Meaning
Total Received	Confirmed incoming transactions
Total Paid	Confirmed outgoing transactions
Balance	Current available/ledger balance according to account model
To Receive	Outstanding receivables

Card	Meaning
To Pay	Outstanding payables
Cash Recorded	Cash transactions recorded in BuildO
Pending/Disputed	Transactions awaiting confirmation/resolution

Payment methods

- Pay by BuildO / integrated digital payment provider.
- Cash.
- Bank transfer or other approved method.
- Payment method must be stored on every transaction.

Cash two-sided confirmation

Payer selects Cash → enters amount → taps I Paid. Recipient gets notification → Received or Dispute. Until recipient confirms, status is pending/disputed as applicable. Once received, both sides' ledgers update and transaction becomes Confirmed.

11. Digital Payments, Wallet/Ledger & Bank Withdrawal

BuildO should separate the internal accounting ledger from regulated payment movement. Use a licensed bank/payment provider integration for actual money movement.

Digital payment flow

Initiate payment → provider authorization → provider callback/webhook → verify transaction → mark Paid/Received → create receipt → update project/order/rental/trip ledger → notify parties.

Cash-to-bank scenario

If a user has recorded/received cash, the user may request a bank transfer/withdrawal only through a supported regulated payment integration. BuildO records the ledger event and provider reference; it should not pretend that cash was digitally moved unless an actual provider transaction succeeds.

Transaction states

Status	Use
Draft	Not submitted
Pending	Awaiting payment/recipient confirmation
Paid	Payer side recorded/verified
Received	Recipient confirmed receipt
Confirmed	Both sides / provider verification complete
Disputed	Recipient or authorized party disputes
Resolved	Dispute resolved with audit record
Failed/Cancelled	Payment failed or cancelled

12. Automatic Financial Linking

No manual duplicate entry should be required when an operational transaction already exists.

Operational event	Automatic financial result
Labour attendance + approved rate	Accrued wage / overtime amount
Labour payment	Labour cost + payer outflow + recipient inflow
Material order	Purchase commitment; on payment, expense/outflow
Rental booking	Rental revenue/expense according to party
Transport trip	Transport income/cost
Customer project payment	Project receipt + receivable reduction
Supplier payment	Payable reduction + company outflow
Refund	Reverse/adjust linked transaction with audit

Project Money view

- Contract Value, Planned Cost, Actual Cost, Received, Receivable, Payable, Cash/Bank activity, estimated margin/profit.
- Filter by date, project, category, person/company, payment method and status.
- Exportable reports and receipts should be supported.

13. Notifications & Automation

Notifications are event-driven. Every notification must have a source event, recipient rule and action/deep link.

Event	Recipient / action
Employment request	Labour → accept/reject
Attendance exception	Supervisor/Owner
Material order	Supplier + buyer
Supplier Ready	Transport network + buyer
Vehicle assigned	Supplier + buyer + driver
Delivery completed	Buyer/project + supplier
Rental due/overdue	Rental owner + renter
Low rental availability	Rental owner
Cash payment awaiting receipt	Recipient
Payment dispute	Payer + recipient + authorized admin
Project stage delay	Project owner/manager
Worker plan range exceeded	Owner; informational only, never block

14. AI Assistant

AI should be an action assistant, not an unrestricted operator. Every action must be permission-checked and high-risk financial actions should require explicit confirmation.

- Answer questions about the user's permitted BuildO data.
- Summarize project progress/costs.
- Warn about labour shortage, late stages, overdue rentals, low stock and pending payments.
- Suggest procurement timing based on project plan and current stock.
- Generate daily/weekly summaries.
- Voice commands can map to allowed actions such as checking attendance, project status, stock or creating a draft request.
- AI must never expose private ratings/comments or financial data outside the user's authorization.

15. Search, Profiles & Trust

Search should be role-aware and location-aware. Results must respect privacy and permissions.

Profile	Public/role-appropriate information
Labour	Name, trade, skills, experience, availability, portfolio/work history allowed by privacy settings, ratings where permitted.
Supplier	Store, categories, products, service area, delivery availability, business verification, ratings/reviews where permitted.
Rental Owner	Rental categories, available inventory, service area, rates/terms, verification.
Driver	Vehicle type/capacity, service area, availability, verification and relevant ratings.
Contractor	Company, trades/services, project history/portfolio, service area and relevant trust information.

16. Suggested Domain/Data Model

This is a conceptual model; engineers may choose PostgreSQL/other technology, but relationships and auditability should be preserved.

Entity	Key relationships
User	has many Roles; many Notifications; financial account/ledger
RoleAssignment	User ↔ Company/Context ↔ Role ↔ permissions
Company	Users, Projects, Products, Assets, Orders, Transactions
Project	Company, Client, Sites, Workers, Jobs, Orders, Rentals, Trips, Transactions
WorkerProfile	User, skills, employment assignments, attendance, work history
Product	Supplier, category, variants, stock records, order lines
Order	Buyer, Supplier, Project, order lines, delivery/trip, payment
RentalAsset/InventoryItem	Rental owner, quantity/asset ID, custody records, maintenance
RentalContract	Customer, project/site, items, dates, rates, payments, returns
Vehicle/Driver	Transport company/owner, trips, availability
Trip	Vehicle/driver, order, pickup/delivery, status, payment
Transaction	From/to, source entity, project, amount, method, status, audit
AuditEvent	Actor, entity, action, old/new values, timestamp

17. API/Event Behaviour Examples

These examples show expected system behaviour; endpoint naming is up to the engineering team.

Command/event	Expected server behaviour
POST attendance/check-in	Validate role + assignment + geofence → create attendance event → notify/record.
POST order	Validate stock/pricing → create order → reserve stock → notify supplier.
PATCH order/ready	Set Ready → trigger transport matching/notifications.
POST trip/accept	Lock assignment → notify supplier/buyer → update order.
POST cash/payment-paid	Create Pending transaction → notify recipient.
POST cash/payment-received	Validate recipient → Confirm transaction → update both ledgers.
POST payment/provider-webhook	Verify provider signature/reference → idempotently update payment.
POST rental/return	Validate custody → update quantities/status → create return record.
POST role-switch	Verify role assignment → issue scoped session/context.
POST AI-action	Check permissions → require confirmation where configured → execute through normal domain service.

18. Security, Audit & Reliability

BuildO will contain employment, location and financial information. Security is a product requirement, not a later add-on.

- Server-side authorization on every protected resource.
- Role/company/project scoped access tokens or equivalent authorization context.
- Encrypt data in transit and at rest.
- Audit changes to money, attendance, stock, rental custody, permissions and order status.
- Idempotency keys for payment/order operations to prevent duplicate submissions.
- Webhook signature verification for payment providers.
- Do not store sensitive bank credentials unless the selected provider explicitly requires and permits a secure tokenized flow.
- GPS data should be collected only for defined operational purposes and according to applicable privacy requirements.
- Provide account/session/device management and suspicious activity alerts.

19. UI / UX Screen Map

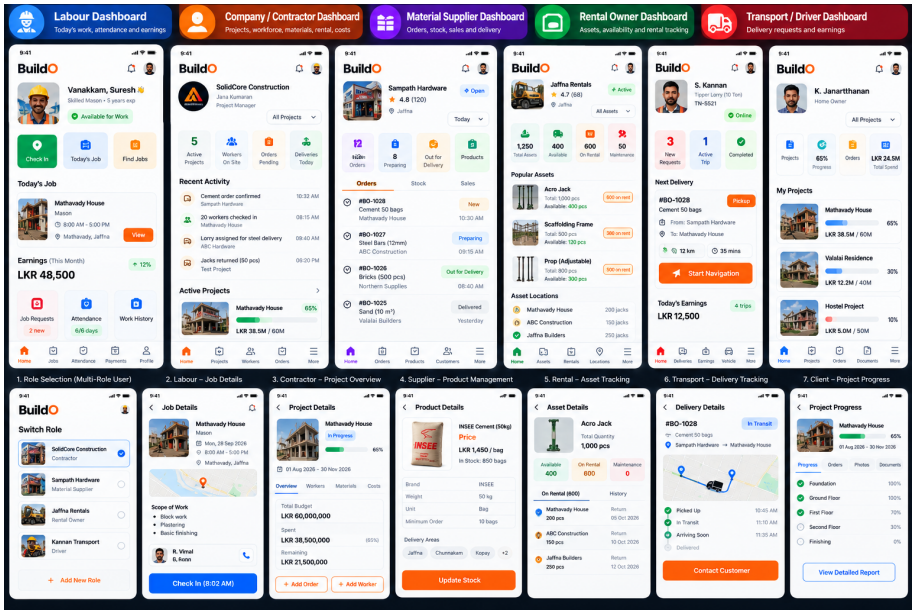
Bottom navigation should be role-dependent. Do not show every module to every user.

Role	Primary navigation example
Labour	Home • Jobs • Attendance • Money • Profile
Contractor	Home • Projects • Workforce • Orders • Money
Supplier	Home • Orders • Catalogue • Deliveries • Money
Rental	Home • Inventory • Rentals • Returns • Money
Driver	Home • Trips • Map • Earnings • Profile
Client	Home • Projects • Orders • Payments • Profile

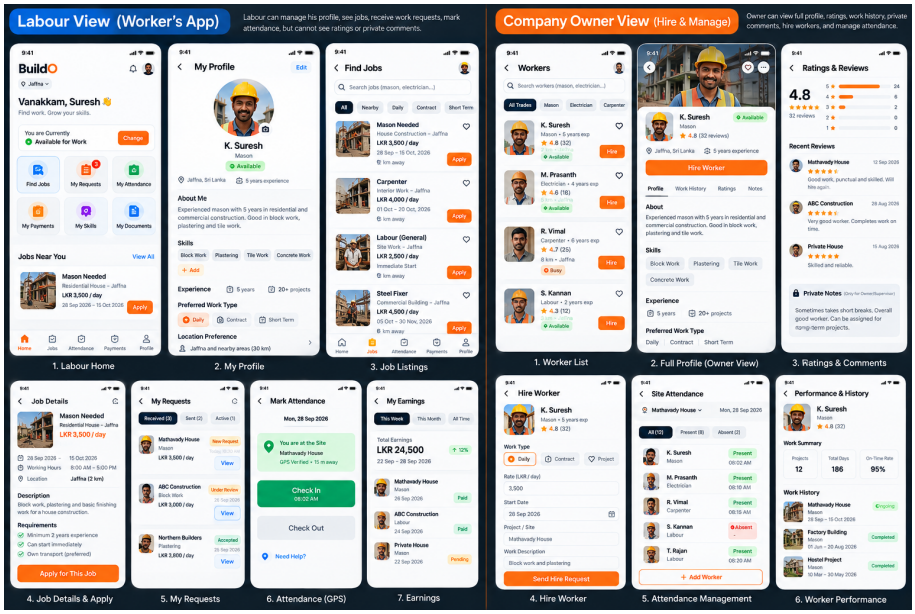
Shared UI principles: clean modern cards, strong status chips, clear primary CTA, map where location matters, timeline for workflows, bottom sheets for quick actions, searchable lists, filters and clear empty/error states.

20. UI Reference Mockups

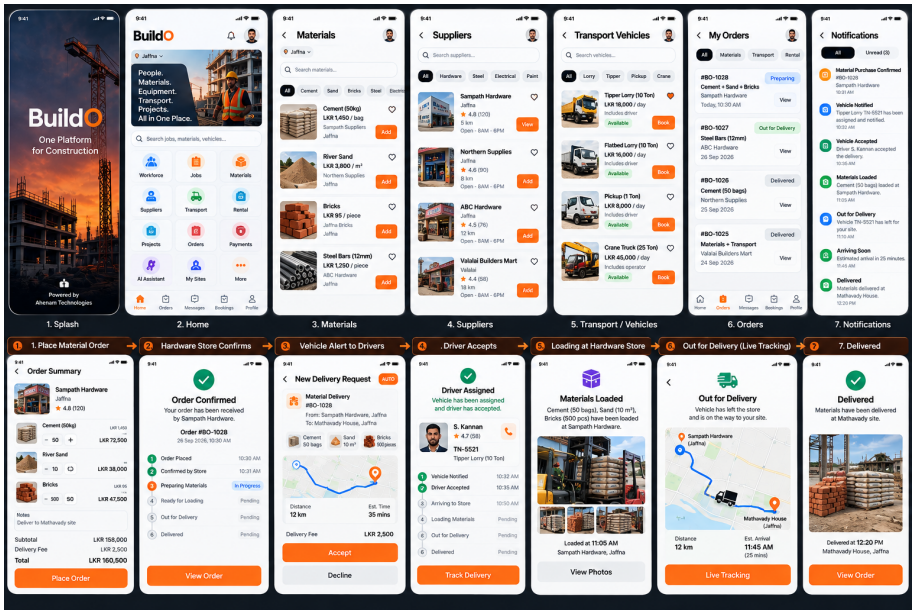
The following generated screens are included as visual references for the intended ecosystem and role separation. They are not final production UI and should not override functional requirements.



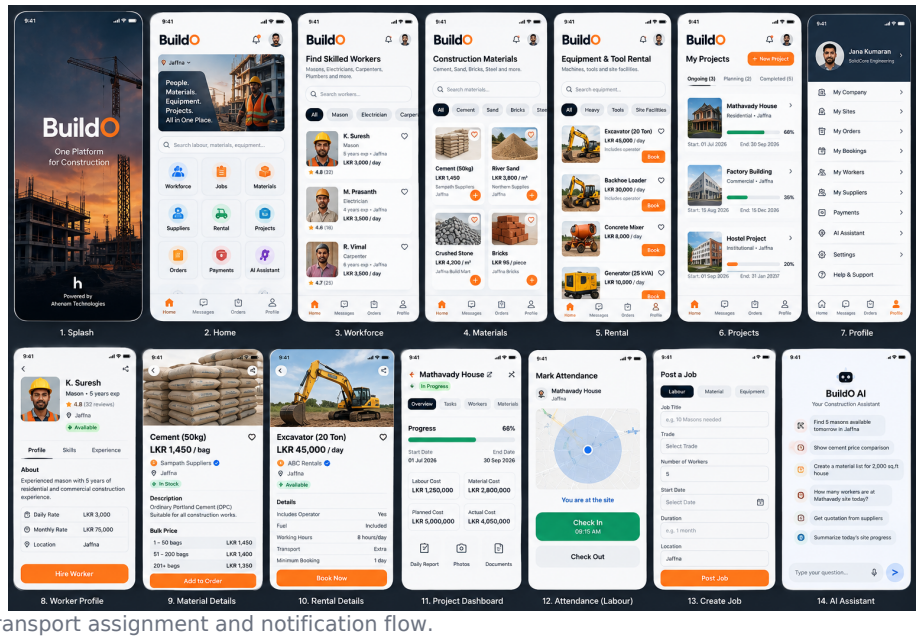
BuildO role dashboards and ecosystem screens.



Labour view vs company owner view.



Materials, supplier, transport and delivery workflow.



Delivery alerts, transport assignment and notification flow.

21. Critical Edge Cases

Engineers should implement explicit states for these cases rather than assuming happy-path behaviour.

- Supplier accepts an order but stock becomes insufficient before dispatch → revalidate stock and notify buyer.
- Two transport drivers accept same delivery → server-side assignment lock allows only one.
- Rental owner has 1,000 units, 600 rented and 50 damaged → available must be 350, not 400.
- Two users try to rent the same individual asset → transaction lock prevents double booking.
- Cash payer marks Paid but recipient disputes → transaction remains Disputed and does not silently become confirmed.
- Payment provider webhook arrives twice → idempotency prevents duplicate ledger entries.
- Worker is outside geofence → labour check-in blocked; authorized supervisor/owner override is audited.
- Worker is removed from project → future attendance actions disabled; history remains.
- Role removed → existing session/context must revalidate permissions.

- Project closed → new operational transactions restricted according to configured close policy; historical data remains.

22. Reports & Exports

Reports should be generated from the same source-of-truth data used by dashboards; avoid separate manually maintained reporting tables where possible.

- Project progress report
- Project planned vs actual cost report
- Labour attendance / overtime report
- Labour payment and dues report
- Material purchase / stock / delivery report
- Supplier transaction report
- Rental inventory/custody/return report
- Transport trip and earnings report
- Money ledger / receivable / payable report
- Daily site report with photos/remarks
- Audit report for sensitive changes

23. Recommended Build Sequence

The entire architecture should be designed now, but development can be phased.

Stage	Release scope
MVP-1	Auth, roles, profiles, permissions, role dashboards, company/project foundation
MVP-2	Workforce, hiring, GPS attendance, work history
MVP-3	Supplier catalogue, materials, stock, orders, quotations
MVP-4	Rental inventory, booking, custody, return, maintenance
MVP-5	Transport/driver, delivery requests, assignment and tracking
MVP-6	Money ledger, cash two-party confirmation, receipts, receivables/payables
MVP-7	Provider-backed digital payments, project financials, reports
MVP-8	AI assistant, automation, voice and advanced analytics

24. Definition of Done — First Production Release

- Each role can complete its core workflow without entering the same data twice.
- Role switch changes permissions and dashboard correctly.
- Project connects workforce, materials, rental, transport and money.
- Supplier catalogue and stock are usable end-to-end.
- Rental inventory quantities remain mathematically correct through booking/return/damage.
- Material Ready event can create a transport request and assignment.
- Delivery status updates order and project.
- Cash payment has payer Paid + recipient Received/Dispute.
- Digital payment is provider-confirmed before final settlement.
- Every financial transaction has an audit trail and unique reference.
- Private labour ratings/comments are inaccessible to labour accounts.
- Attendance respects GPS and supervisor/owner override permissions.
- Notifications deep-link to the relevant action.
- Reports reconcile to underlying transactions.
- Critical operations are protected against duplicate requests and race conditions.

Final engineering principle

Do not build BuildO as a collection of disconnected screens. Build a connected domain system where one action automatically updates the relevant operational record, notification, project status, inventory and financial ledger.

BuildO = One connected construction operating network.